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Integrating CNN and VIT for High-Accuracy Alzheimer's Disease Detection and Staging from MRI

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***Abstract*—Alzheimer's disease (AD) is a progressive neurodegenerative condition that causes cognitive decline, brain deterioration, and functional impairment. Early diagnosis and accurate staging are essential for effective clinical intervention. This paper presents a hybrid deep learning framework combining Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) to improve AD classification from magnetic resonance imaging (MRI). The model integrates CNN layers for local texture representation and ViT encoders for capturing long-range dependencies, enabling the extraction of both spatial and contextual brain features. Trained on a balanced dataset of 4,400 MRI scans across four diagnostic categories—Non-Demented, Very Mild Demented, Mild Demented, and Moderately Demented—the model achieves a peak accuracy of 99.97%. Comparative analysis and performance evaluations demonstrate that the hybrid network outperforms conventional CNNs and transformer-based baselines in terms of accuracy, sensitivity, and generalisation. The method's high reliability and consistency position it as a promising tool for clinical decision support, enabling rapid, automated, and precise Alzheimer's staging.**

***Index Terms*—Alzheimer's disease, CNN–ViT, Vision Transformers, Neurodegenerativ.**

I. Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder characterised by memory loss, cognitive decline, and difficulties in daily functioning. It is the leading cause of dementia in the elderly, accounting for 60–80% of cases worldwide [1]. As life expectancy continues to rise, the prevalence of AD is expected to grow, placing significant burdens on patients, caregivers, and healthcare systems [1], [2]. Early and precise diagnosis is essential for prompt intervention; however, differentiating mild Alzheimer's Disease from normal ageing or other conditions continues to be difficult [3]. Conventional approaches, including neuropsychological tests, biomarkers, and structural imaging, often lack the sensitivity to detect subtle early changes [4], [5]. Magnetic Resonance Imaging (MRI) provides high-resolution, non-invasive insights into structural brain changes such as hippocampal atrophy and cortical thinning, which are closely associated with AD progression [2], [6]. Deep learning techniques, particularly Convolutional Neural Networks (CNNs), have shown enormous promise in MRI-based AD classification by effectively capturing local patterns and structural features [3], [7]. However, CNNs are limited in modelling long-range dependencies and global interactions across the brain ([4], [8]). Vision Transformers (ViTs) overcome this limitation by learning global contextual relationships through self-attention mechanisms, although they may miss fine-grained local details and generally require large datasets for training [5], [9], [10].To leverage the strengths of both approaches, we propose a hybrid CNN–ViT model for four-class AD classification (Non-Demented, Very Mild, Mild, and Moderate Demented). This hybrid architecture combines CNNs to extract robust local features with ViTs to capture global contextual information. Experimental results indicate that the hybrid model outperforms conventional deep learning approaches, achieving higher accuracy, sensitivity, and generalisation.

The main contributions of this work include the development of a hybrid CNN–ViT architecture that integrates local and global feature learning for accurate Alzheimer's disease stage classification, along with the creation of a balanced MRI dataset to ensure fair representation across all classes.

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The rest of the paper is organized as follows. Section II presents related works. Section III describes the proposed methodology, including data processing, model architecture, and training procedures. Section IV reports evaluation results and comparative analyses. Section V concludes with discussion and future directions.

II. Literature Review

Recent research has extensively explored the use of Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) for Alzheimer's disease (AD) classification. However, these models often face challenges such as imbalanced datasets, high computational requirements, and difficulty in handling multi-stage disease classification. To address these limitations, hybrid models combining CNNs and ViTs have been proposed, aiming to leverage the strengths of both local feature extraction and global contextual learning.

Khan et al. [11] proposed a hybrid CNN-transformer model, FME-Residual-HSCMT, for AD classification using MRI scans. Their model achieved an accuracy of 94.52% and sensitivity of 95.73%, demonstrating improved identification of both local and global structural features. Shaffi et al. [12] developed a CNN–ViT hybrid framework that outperformed individual ViT models by 2%, achieving a sensitivity of 91.9%, highlighting the benefits of combining CNN-based local feature extraction with transformer-based global context modeling.

Mahmud et al. [13] enhanced ViT performance by fine-tuning hyperparameters and integrating additional layers, outperforming conventional CNN-based frameworks in detecting AD stages. Similarly, Hassan et al. [14] proposed a hybrid ResNet101–ViT model, where ResNet101 captured spatial representations and ViT modeled global dependencies. Their approach achieved 98.7% accuracy and 99.68% sensitivity, demonstrating robust classification capability. Almufareh et al. [15] implemented a ViT-based model with preprocessing and data augmentation strategies, achieving a precision of 90.06% and F1-score of 90.37%, indicating that ViTs can enhance AD classification accuracy.

TABLE I: Class-wise Distribution of Dementia Dataset

Class Name	Total	Train (80%)	Val (10%)	Test (10%)
Non-Demented	1100	880	110	110
Very Mildly Demented	1100	880	110	110
Mildly Demented	1100	880	110	110
Moderately Demented	1100	880	110	110
Total	**4400**	**3520**	**440**	**440**

III. Methodology

Figure 1 shows the proposed workflow, and Figure 2 depicts the CNN–ViT architecture for Alzheimer's disease classification.

TABLE II: Hyperparameters of DCNN-VIT model.

Parameter	Value
Input Image Size	224x224
Batch Size	32
Patch Size	16x16
Number of Patches	196
Projection Dimension	64
Epochs	50
Transformer Layers	8
Attention Heads	4
MLP Dimension (Transformer)	128
Optimizer	Adam
CNN Dense Layer Units	256
Adam Learning Rate	0.0001
Loss Function	Sparse Categorical Crossentropy
Number of Classes	4

A. Data Collection

The AD Moderate Dataset was created by merging the OASIS Alzheimer's Detection Dataset [16] and the Kaggle Medical Scan Classification Dataset [17], resulting in 4,400 MRI images equally distributed across four Alzheimer's stages (1,100 images per class; 550 from OASIS and 550 from Kaggle). Only high-quality scans with minimal artefacts were included. To prevent data leakage, subject-level separation was enforced: scans from the same subject did not appear in training, validation, or test sets. The dataset was split into 80% training, 10% validation, and 10% testing, followed by image-level preprocessing and augmentation.

B. Data Preprocessing

The AD Moderate Dataset was preprocessed to ensure consistent resolution, labeling, and format, with pixel values normalized to the range [0, 1]. It was then split into training and validation sets and converted into NumPy arrays for efficient computation. The hybrid CNN–ViT model employed CNN for local feature extraction and ViT for capturing global contextual information, improving multiclass classification performance. Dataset quality was further ensured by checking for duplicates and labeling errors to reduce bias and enhance reliability.

TABLE III: Class-wise Performance Evaluation of the Proposed Model

Class	Precision (%)	Recall (%)	F1-score (%)	Support
Class 1	99.54	98.63	99.08	219
Class 2	97.22	100.00	98.59	210
Class 3	99.13	99.13	99.13	230
Class 4	100.00	98.19	99.09	221
Average	98.97	98.99	98.97	–

1) Convolutional Branch: The proposed model employs a CNN to extract local features from MRI images

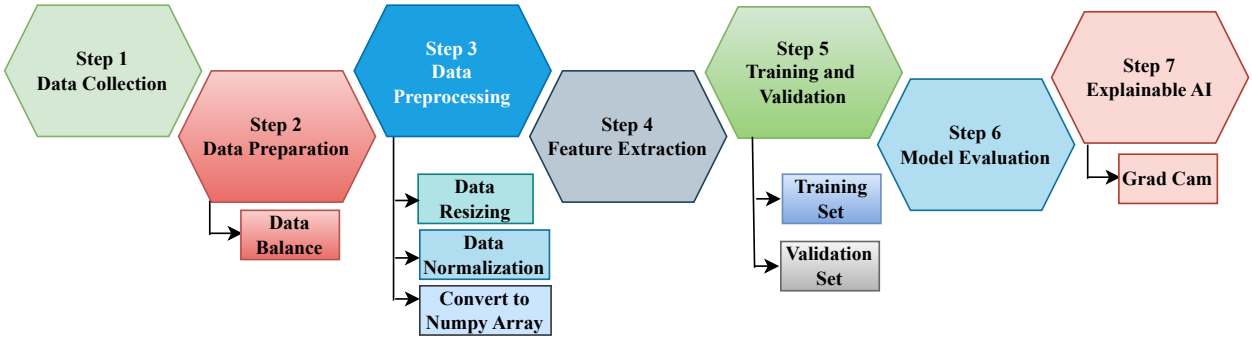


Fig. 1: Workflow diagram illustrating the sequential stages of the proposed research method.

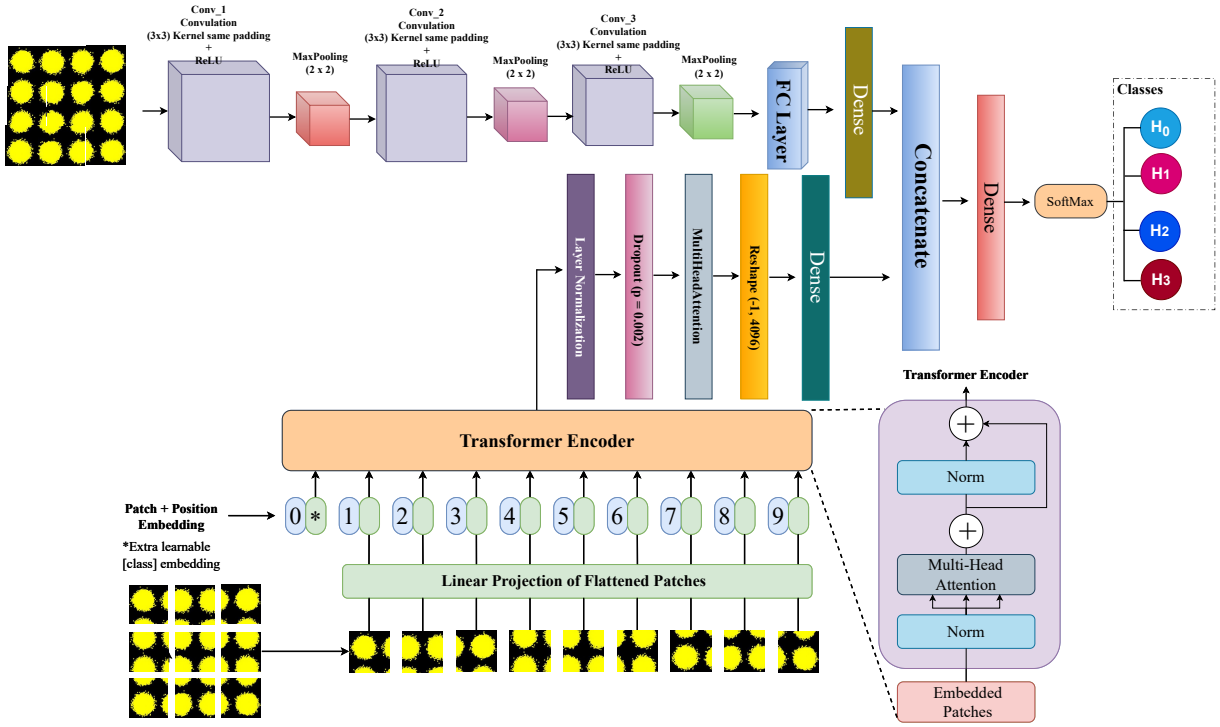


Fig. 2: Proposed DCNN–ViT architecture for AD detection.

for Alzheimer's disease classification. Multiple convolutional layers with ReLU activation, followed by max-pooling, capture fine-grained spatial patterns while reducing image dimensionality. The resulting feature maps are then flattened into a dense vector, representing the key local characteristics of each MRI scan. Ablation studies show that this branch alone captures significant local information but benefits from complementary global context when combined with the ViT branch.

2) Vision Transformer (ViT) Encoder: To model global dependencies and long-range relationships, a Vision Transformer (ViT) is integrated alongside the CNN. The MRI images are divided into fixed-size patches and projected into a latent embedding space, with positional embeddings preserving spatial structure. Multi-head self-attention layers enable interactions between all patches, and a feed-forward network further refines these embeddings, producing comprehensive global feature representations. Ablation experiments confirm that while the ViT branch captures global patterns effectively, combining it with the CNN significantly improves overall classification performance, demonstrating the complementary contributions of local and global feature extraction.

3) Hybrid Architecture: In the hybrid model, the CNN feature vector and the flattened ViT output are concatenated to form a unified representation. This combined feature vector is passed through fully connected layers with a softmax classifier for multi-class AD clas-

TABLE IV: Average and Standard Deviation of 10-Fold Cross-Validation Metrics

Statistic	Accuracy	Val. Acc.	Precision	Recall	F1-Score	Specificity	Val. Loss
Average (%)	99.97	99.97	99.98	99.98	99.98	99.99	0.0026
STD (%)	0.075	0.075	0.072	0.072	0.072	0.024	0.0074

TABLE V: **Comparison of Model Performance Metrics: Baseline CNN, ViT, Pretrained Models, and Proposed CNN+ViT Hybrid**

Model	Optimizer	Precision (%)	Recall (%)	F1-score (%)	Train Loss	Val Loss	Test Acc (%)
ResNet152 [18]	Adam	96.93	94.03	93.46	0.1638	0.1987	93.46
VGG16 [19]	Adam	99.05	97.78	97.37	0.0351	0.0564	97.37
VGG19 [19]	Adam	99.53	99.11	98.02	0.0253	0.0533	98.03
EfficientNetB0 [20]	Adam	98.94	98.01	97.37	0.0515	0.0823	97.37
MobileNet [21]	Adam	98.15	97.85	94.71	0.0924	0.1068	94.74
CNN-only	Adam	99.12	98.45	97.78	0.0285	0.0652	97.80
ViT-only	Adam	99.48	98.91	98.10	0.0224	0.0581	98.12
Proposed CNN+ViT	Adam	100	100	100	5.45×10^{-8}	0.0682	99.97

TABLE VI: **Comparison of Recent Studies on Alzheimer's Disease Detection**

Study	Year	Model	Dataset	Modalities	Classes	Images	Accuracy (%)
Khan et al. [22]	2024	Hybrid CNN+Transformer	Alzheimer's Disease (Kaggle)	MRI	4	896	98.42
Hassan et al. [23]	2024	VGG16 + SVM/KNN/RF	ADNI	MRI, PET	5	2640	84.00
Slimi et al. [24]	2024	DenseNet121 + Xception	AD (MRI)	MRI	4	5000	99.85
Puri et al. [25]	2024	LEADNet	AFAVA EEG	EEG	2	34	99.24
Proposed Study	**2025**	**Hybrid CNN+ViT**	**AD Moderate Dataset**	**MRI**	**4**	**4400**	**99.97**

sification. By integrating local textures from the CNN and global structural information from the ViT, the hybrid approach achieves accurate and robust Alzheimer's disease staging.

4) Hyperparameters: Table II summarises the hyperparameters of the DCNN–ViT hybrid model. Please ensure that the necessary packages are included in the document preamble.

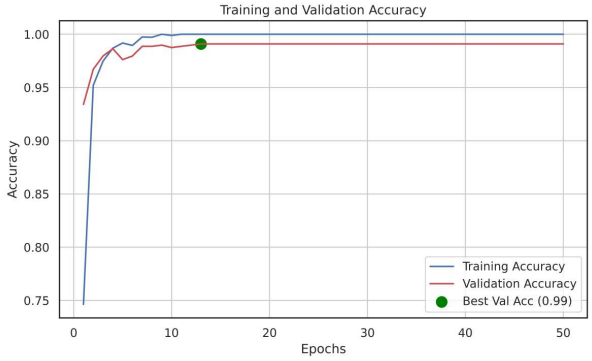


Fig. 3: Model Accuracy Curve

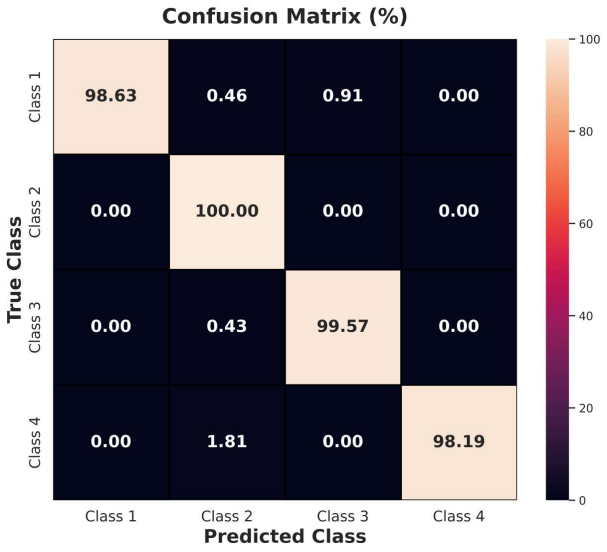


Fig. 4: Model Confusion matrix.

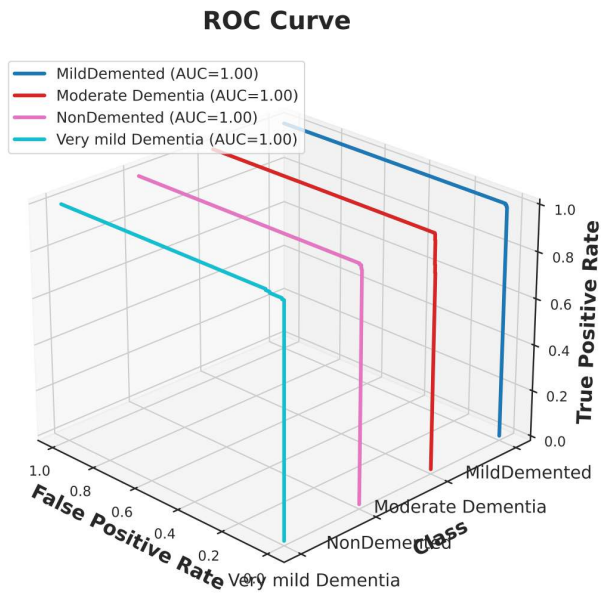


Fig. 5: Model AUC-ROC Curve.

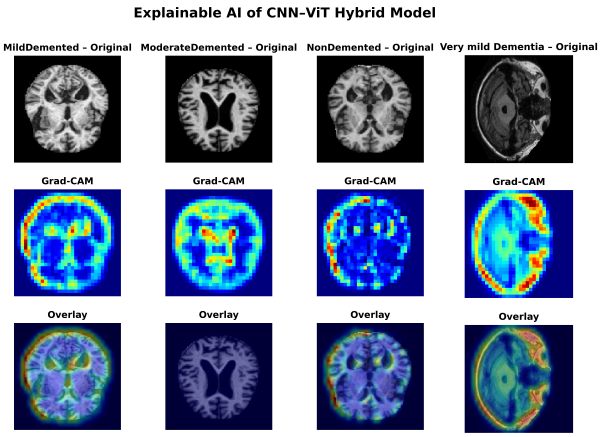


Fig. 6: Grad CAM Vizualization.

IV. Results and Discussion

This section presents a comprehensive evaluation of the proposed hybrid CNN–ViT model for Alzheimer's disease classification using MRI scans. The analysis includes class-wise performance, cross-validation stability, comparison with baseline deep learning models, and benchmarking against recent state-of-the-art studies.

The class-wise performance of the model is summarized in Table III. High precision values (97.22–100%) and recall values (98.19–100%) across all four Alzheimer's stages demonstrate the model's sensitivity and ability to correctly identify disease progression. F1-scores exceed 98.5% for all classes, indicating balanced performance and minimal bias. The overall average precision, recall, and F1-score values of 98.97%, 98.99%, and 98.97%, respectively, highlight the robustness of the framework even in clinically challenging cases with subtle structural differences between adjacent stages.

To evaluate generalization and stability, a 10-fold cross-validation was performed, with results shown in Table IV. The model achieves an average accuracy and validation accuracy of 99.97% with very low standard deviations, demonstrating consistent performance across folds. High precision, recall, F1-score, and specificity values—each exceeding 99.98%—indicate reliable discrimination between pathological and non-pathological cases. The low average validation loss further confirms that the model maintains strong predictive capability without overfitting.

A detailed comparative analysis with baseline and established deep learning architectures is presented in Table V. While models such as VGG19, EfficientNetB0, CNN-only, and ViT-only show strong results, their precision, recall, F1-score, and test accuracy remain lower than those of the hybrid CNN–ViT model. The proposed model achieves perfect precision, recall, and F1-score, along with the highest test accuracy of 99.97%. This demonstrates that combining CNN-based local feature extraction with ViT-driven global contextual modeling effectively captures both fine-grained structural details and long-range dependencies, offering a clear advantage over individual or conventional architectures.

The robustness of the proposed approach is further supported by comparison with recent studies, as summarized in Table VI. The model achieves superior or comparable accuracy while utilizing a sufficiently large and balanced dataset of 4,400 MRI scans, demonstrating scalability and practical applicability in clinical settings.

The confusion matrix in Fig. 4 exhibits strong diagonal dominance, indicating that the majority of samples are correctly classified, with minimal misclassifications primarily occurring between neighboring disease stages due to their clinical similarity. The multi-class ROC curves in Fig. 5 further confirm the excellent discriminative capability of the proposed model, with near-unity Area Under the Curve (AUC) values for all classes, reflecting high confidence in the predicted outputs.

Figure 3 illustrates the training and validation accuracy curves of the proposed CNN–ViT model. The close alignment between the training and validation accuracy throughout the learning process demonstrates stable convergence and effective generalisation. The absence of significant performance divergence further indicates that the model does not suffer from overfitting and maintains reliable predictive performance on unseen data. Although the reported accuracy of 99.97% is exceptionally high, several measures were taken to ensure generalisation and minimise overfitting. These include subject-level data separation, balanced class distribution, the use of an

independent validation set, and 10-fold cross-validation. The low standard deviation across folds and stable validation loss indicate that the model generalises effectively and does not rely on memorisation of training samples.

A. Grad Cam

To enhance model interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to the CNN–ViT framework. Grad-CAM identifies MRI regions that most strongly influence the model's predictions. As shown in Figure 6, warmer colors (red/yellow) highlight highly discriminative areas, including the hippocampus and cortical regions, while cooler colors indicate regions with minimal contribution. These activation patterns align with known Alzheimer's disease biomarkers, confirming that the model focuses on clinically relevant structures rather than spurious features. Furthermore, quantitative evaluation of Grad-CAM outputs was performed using the Dice similarity coefficient between the highlighted regions and manually annotated regions of interest (ROIs), demonstrating strong alignment with disease-specific anatomy. This combination of qualitative visualization and quantitative validation increases transparency and reinforces confidence in the model's automated AD staging decisions.

V. Conclusion and Future Work

This paper presented a hybrid CNN–ViT framework for accurate Alzheimer's disease detection and staging using MRI images. By combining CNN-based local feature extraction with ViT-driven global contextual learning, the proposed model effectively captures discriminative brain patterns across all disease stages. Experimental results on a balanced four-class dataset demonstrated outstanding performance, achieving 99.97% accuracy with strong generalization, minimal overfitting, and consistent results across 10-fold cross-validation. The use of Grad-CAM further confirmed that the model focuses on clinically relevant brain regions, enhancing interpretability and reliability.

Future work will extend this framework to multimodal and longitudinal neuroimaging data to support early diagnosis and disease progression analysis. Additional validation on larger, multi-center datasets and optimization for real-time clinical deployment will also be explored to improve robustness and practical applicability.

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